INCH-POUND
MIL-DTL-70745A (AR)
8 December 2011
SUPERSEDING
MIL-S-70745 (AR)
w/AMENDMENT 3
28 February 1995

DETAIL SPECIFICATION

STRIP, AMMUNITION, METALLIC – 30 MM

Reactivated after 8 December 2011 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the US Army Armament, Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense within the distribution limitations noted at the bottom of the page.

1. SCOPE

1.1 Scope. This specification contains the requirements for Strip, Ammunition, Metallic – 30 Millimeter hereafter referred to as the strip (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000, or ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

11825935 - Cartridge, 30 mm - Dummy M848
12902719 - Strip, Ammunition, Metallic, 30 mm
12910289 - Strip Packed (30 mm Metallic Strips) 30 mm Cartridges

(Copies of these drawings may be requested online at pica.drawing.request@conus.army.mil or from US Army ARDEC, RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

(Copies of these documents are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or http://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and Table II.

3.1.2 Conformance. For each lot a sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Metallic 30 MM ammunition strip. The strip shall comply with all requirements in accordance with 12902719.

3.3 Push out. The individual force required to disassemble the nose end of a lightweight 30 mm cartridge family (see 6.8) on 11825935 from the ammunition strip shall be not less than 5 pounds nor greater than 30 pounds. The base of the cartridge does not have to be retained in the rear clip after the push out test.

3.4 Functioning. The strip shall be capable of being assembled with the lightweight 30 mm cartridges family (see 6.8) and cycled through the AH-64 Apache Uploader/Downloader system and the AH-64D Apache Side Loader system without binding, cracking, breaking, separating or causing a malfunction of the loading systems.

3.5 Corrosion resistance. The strip shall be finished in accordance with 12902719. When exposed to a 5 percent solution of atomized salt spray for 24 hours, the strip shall exhibit no evidence of corrosion.

3.6 Burrs and sharp surfaces. All surfaces of the strip shall be free of any burr or sharp surfaces which could cause removal of metal from any part of the cartridge.

3.7 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.8 Workmanship. The parts shall be free of defects such as folds, wrinkles, buckles, splits, severe scratches, cuts, dents, perforations, sharp edges, unblended radii, surface defects. All parts and assemblies shall be clean and free of burrs, chips, dirt, grease, oil, corrosion products, and foreign matter.

3.8.1 Metal defects. The strip shall be free of blisters, seams, laminations and porosity. Cracks or tears are not permitted per Figure 3 principal areas.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Methods of Verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		Classes of Verification A – First article B – Conformance						
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Method
		1	2	3	4	A	B	
3.1.1	First article			X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2	Metallic 30mm ammunition strip			X	X	X	X	4.3.2.1
3.3	Push out				X	X	X	4.4.1
3.4	Functioning				X	X		4.4.2
3.5	Corrosion resistance			X	X	X		4.4.3
3.6	Burrs and sharp surfaces			X		X	X	4.4.4
3.7	Ammunition lot numbering			X		X	X	4.6
3.8	Workmanship			X		X	X	4.4.5
3.8.1	Metal defects			X		X	X	4.4.5.1

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. When specified, samples of the strip and components shall be subjected to first article inspection in accordance with Table II.

4.2.1 First article rejection. Unless otherwise specified, if any item of the sample fails to comply with the first article requirements, the first article shall be rejected.

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TABLE II. First article inspection.

Examination or Test	No. of Sample Units (bulk strips)	Requirement Paragraph	Inspection Method
Examination for defects Metallic 30mm ammunition strip	252 1/	3.2	4.3.2.1
Push out	20	3.3	4.4.1
Functioning	200	3.4	4.4.2
Corrosion resistance	32	3.5	4.4.3
NOTE: 1/ Strips used for examination for defects may be used for push out, functioning, and/or corrosion resistance tests.			

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916.

4.3.2 Classification of characteristics.

a. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in the conformance examination/test paragraph and MIL-STD-1916. Unless specified otherwise, Verification Level VII in addition to 100% inspection shall be used for all characteristics defined as critical. Verification Level IV shall be used for all characteristics defined as majors and Verification Level II for all minor characteristics.

b. Conformance inspection shall be performed in accordance with 4.3.2.1.

c. The Acceptance Inspection Equipment (AIE) shall be in accordance with MIL-A-48078. AIE shall be capable of repeatable measurements, by various experience inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristics being inspected.

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4.3.2.1	Metallic 30 mm ammunition strip.			Drawing Number 12902719
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
CRITICAL	None defined			
MAJOR				
101	Centerline to centerline of clips (10 equal spaces)	Level IV	3.2	AIE
102	Hole centerline to edge (2 places)	Level IV	3.2	AIE
103	Width of strip (11 places)	Level IV	3.2	AIE
104	Hole centerline to outer edge of clip (11 places)	Level IV	3.2	AIE
105	Hole centerline to far clip inner edge (11 places)	Level IV	3.2	AIE
106	Hole centerline distance (11 places)	Level IV	3.2	AIE
107	Hole centerline to near clip inner edge (11 places)	Level IV	3.2	AIE
108	Hole centerline to flange edge (11 places)	Level IV	3.2	AIE
109	Material thickness	Level IV	3.2	AIE
110	Push out	N=20 1/	3.3	4.4.1
111	Burrs and sharp surfaces	Level IV	3.6	Visual
112	Diameter of holes (22 places)	Level IV	3.2	AIE
MINOR				
201	Retaining tab to centerline spread (44 places)	Level II	3.2	AIE
202	Location to angle center (22 places)	Level II	3.2	AIE
203	Clip bend angle (44 places)	Level II	3.2	AIE
204	Clip bend radius (44 places)	Level II	3.2	AIE
205	Clip base width	Level II	3.2	AIE
206	Heel line angle location at base of clip (22 places)	Level II	3.2	AIE
207	Clip closing angle (44 places)	Level II	3.2	AIE
208	Clip base width (22 places)	Level II	3.2	AIE
209	Clip base to centerline (22 places)	Level II	3.2	AIE
210	Warpage in free space	Level II	3.2	AIE
211	Workmanship	Level II	3.8	Visual and 4.4.5

NOTE:

1/ Accept on zero, reject on one.

4.4 Methods of inspection.

4.4.1 Push out. A strip (12902719) shall be assembled with M848 dummy cartridges (11825935) in accordance with 12910289 and shall be placed into the test fixture (see Figure 1). In order to test both the front and aft clips, half of the strip samples shall be assembled with M848 dummy cartridges (11825935) and tested with the strip lot number oriented on the right side and the other half of the strips with the lot number on the left side. There shall be no movement of the

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dummy cartridges in the strip prior to assembling in the test fixture. A gradual force (without impact loading) shall then be applied to a cartridge until the nose end has completely separated from the strip. Measurement of the maximum push out force shall be made for compliance with the requirements of 3.3. The strip shall be rigidly supported to have the least amount of movement when the push out force is being applied. The pushed out cartridge shall be returned to the strip and the above procedure repeated for all remaining cartridges regardless of the position of a cartridge in the strip. When pushing out the cartridges at either end of the strip the tabs at the end of the strip shall be rigidly supported to have the least amount of movement or bending of the strip when the push out force is being applied.

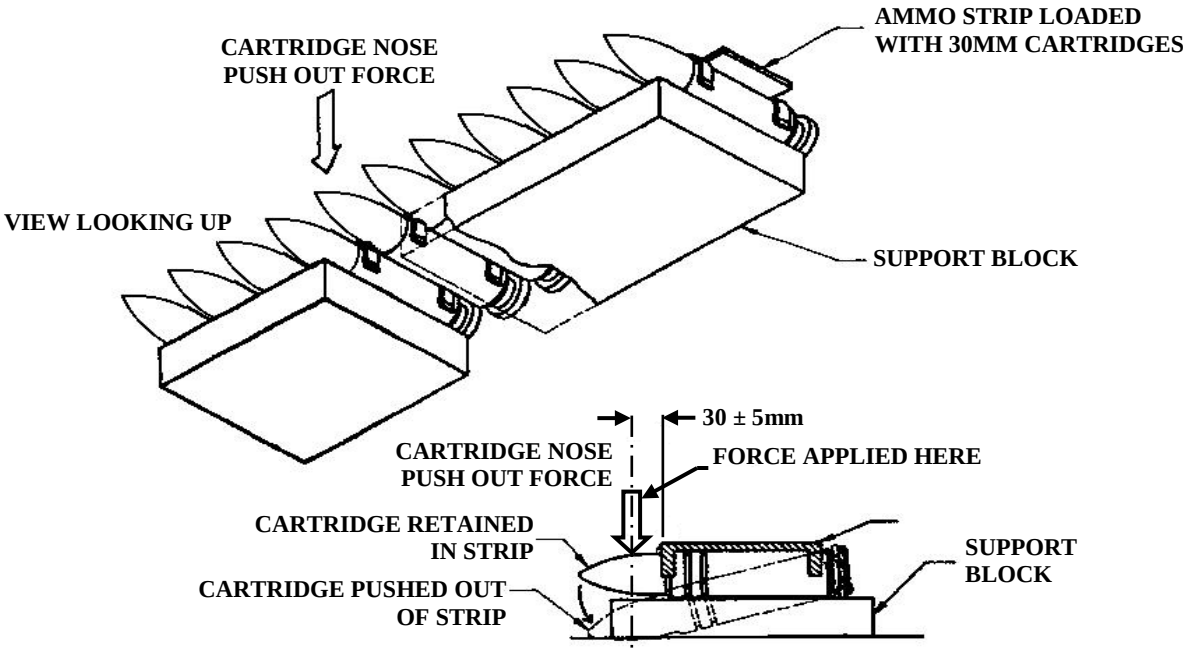


FIGURE 1. Push out force test set-up.

4.4.2 Functioning. A strip (12902719) shall be assembled with M848 dummy cartridges (11825935) in accordance with 12910289 and the strip shall be fed through the AH-64D Apache Side Loader system. Then the ammunition shall be downloaded back into the strip using the Side Loader system and subsequently feed into the AH-64 Apache Uploader/Downloader. This test shall be conducted at range ambient temperature. Failure of strip to retain cartridges throughout uploading and/or downloading and meet the requirements of 3.4 shall be cause for rejection.

4.4.3 Corrosion resistance. The test shall be conducted in accordance with ASTM B117 and 3.5. Tested parts shall not be returned to the lot.

4.4.4 Burrs and sharp surfaces. Observations shall be made for compliance with the requirements of 3.6.

4.4.5 Workmanship. The metallic strip shall be examined for compliance with the workmanship requirements as stated in 3.8 and 3.8.1.

4.4.5.1 Metal defects - cracks or tears.

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4.4.5.1.1 Cracks or tears in principal areas. Any visible crack or tear in the strip extending through the stock thickness in depth, or a distance in length equal to or greater than the stock thickness shall be classified as a defect (see Figure 2 and Figure 3 Notes 1 and 2). Cracks or tears are not permitted in radii where the clips meet the base part of the strip (Figure 3 Note 2)

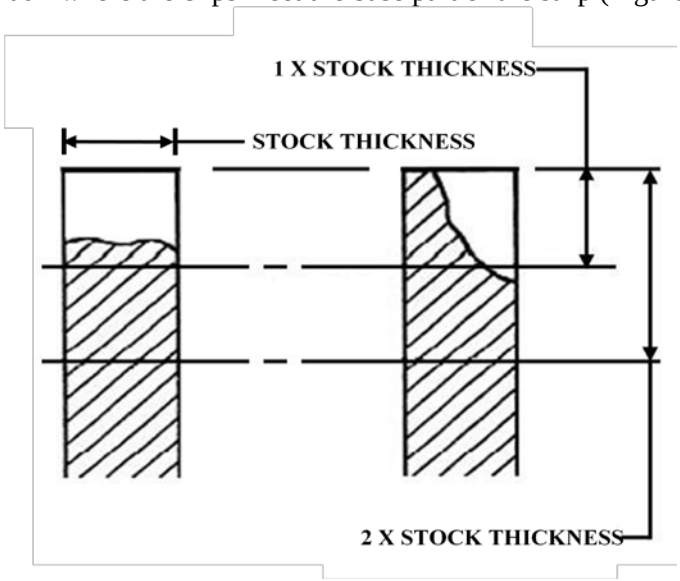
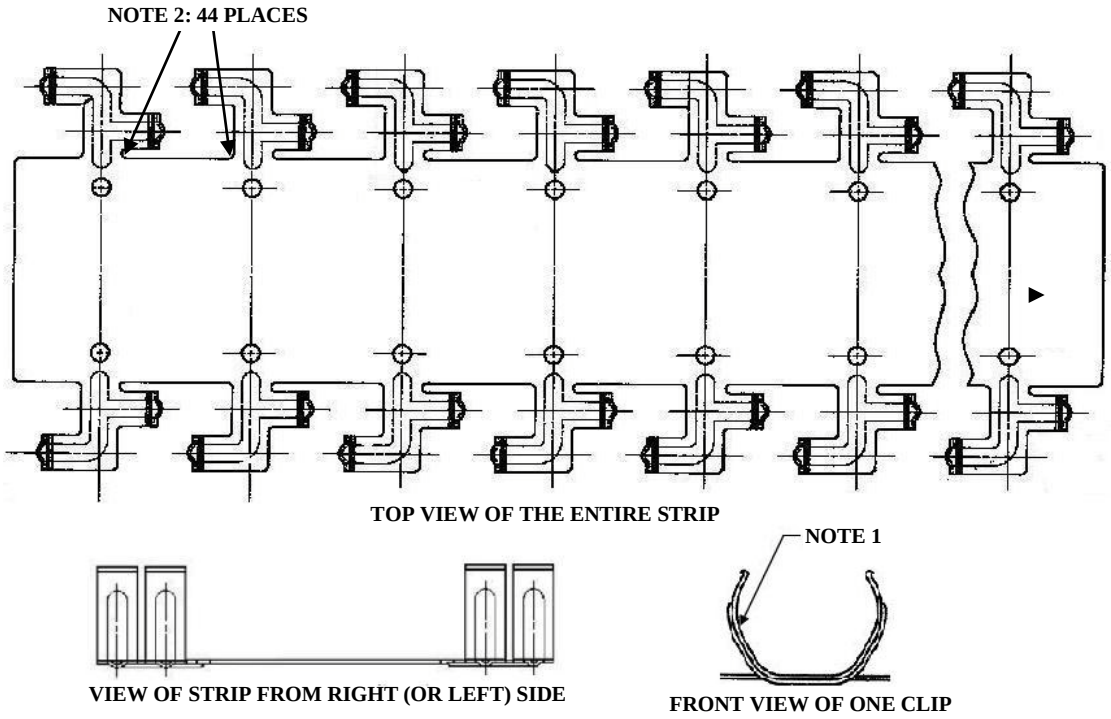


FIGURE 2. Example of cracks and tears in principal areas.



Note 1: Entire area of clip is designated as a principal area.
Note 2: No crack or tear allowed in these 44 radii.

FIGURE 3. Definition of principal areas.

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4.4.5.1.2 Cracks or tears in non-principal areas. All areas not specifically designated in Figure 3 as principal areas are non-principal areas. Any visible crack or tear in the strip extending through the stock thickness in depth, and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length shall be classified as a defect (Figure 4).

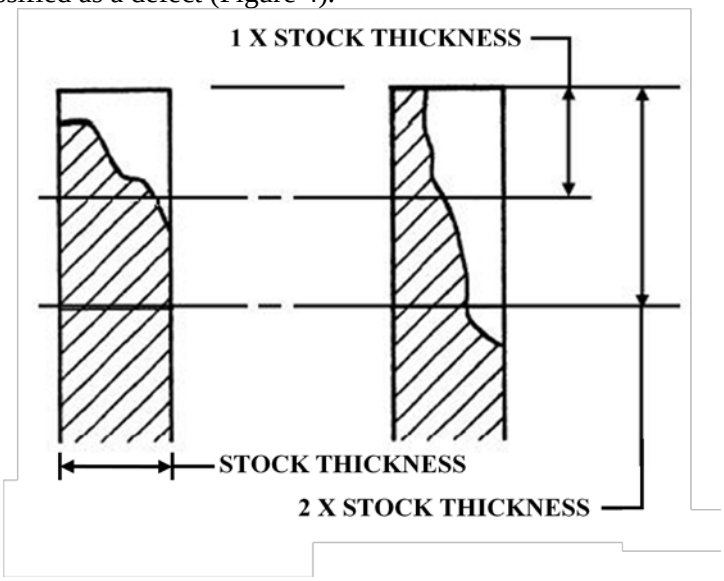


FIGURE 4. Example of cracks and tears in non-principal areas.

4.6 Ammunition lot numbering. Visually verify that ammunition lot number has been applied to each lot as described in MIL-STD-1168.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers fabrication of the Strip, Ammunition, Metallic - 30 mm for use in packing and loading LW30 mm ammunition. There is no known commercial use or application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and all reference documentation cited in this specification (see 2.2.1).

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b. Requirements for submission of first article: A first article sample, either in part or complete (Table II), may be required for the commencement of production after the award of a new contract, a change in production venue, a process change for any part or subassembly, or after a production stoppage in excess of 90 days.

c. Requirement for submission of inspection equipment designs.

d. Requirement and provisions for submission of test data as required.

e. Provisions for the inclusion of MIL-STD-1168, Ammunition Data Cards on DD Form 1423, Contract Data Requirement List.

f. Conformance inspection, other than specified in Section 4 of this specification.

g. Applicable National Stock Number.

h. Lists of drawings, publications and specifications, showing applicable revision dates.

i. Certificate of conformance for each lot or shipment of product, if applicable.

j. Place of inspection, if not at place of manufacture.

k. Government Furnished Material or Equipment.

l. Packaging requirements.

6.3 Process deviation. A process deviation defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Visual defects standards. Components and assemblies which have been rejected during initial production should be segregated and visual standards established jointly by representatives of the Government and the contractor. Subsequent application of these standards for acceptance inspection should be specified in the contract or purchase order.

6.5 Automatic acceptance inspection equipment (AAIE) and acceptance inspection equipment (AIE). Provision concerning the AAIE and AIE used to verify the requirements of this specification should be specified in the contract if applicable.

6.6 Submission of inspection equipment designs for approval. Submit copies of designs as required to: Commander, US Army ARDEC, Attn: RDAR-QEM-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.7 Submission of alternative quality conformance provisions. All proposed alternative quality conformance provisions should be submitted for evaluation/approval as directed.

6.8 Lightweight 30 mm cartridge family. The Lightweight 30 mm cartridge family consistence of but is not limited to the following: M789, M788 and M848.

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6.9 Subject term (key words) listing.

Push out
Cracks or tears

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
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